Only one surface it to be polished and maintained. Support can be given from the entire back of the mirror whereas it is given only at the rim for lens. Mirror weigh much less compared to lens. But, the one obvious problem with a reflecting telescope is that the objective mirror would focus the light inside the telescope tube. One must have an eye piece inside the tube obstructing some light. This problem could also be overcome by introducing a secondary mirror which would take the light outside the tube for view as shown in the Figure 7.42.

## 7.6.6 Spectrometer

The spectrometer is an optical instrument used to analise the spectra of different sources of light, to measure the wavelength of different colours and to measure the refractive indices of materials of prisms. It is shown in Figure 7.43. It basically consists of three parts namely (i) collimator, (ii) prism table and (iii) telescope.



**Figure 7.43** Spectrometer

### *(i) Collimator*

The collimator is used for producing parallel beam of light. It has a convex lens and a vertical slit of adjustable width which faces the source. The position of slit can be adjusted so that it is kept at the focus of the lens. The collimator is rigidly fixed to the base.

### *(ii) Prism table*

The prism table is used for mounting the prism, grating etc. It consists of two circular discs provided with three levelling screws. It can be rotated and its position can be read from two verniers $V_1$ and $V_2$ . The prism table can be fixed at any desired height.

### *(iii) Telescope*

The telescope is an astronomical type. It consists of an eyepiece provided with cross wires at one end and an objective at its other end. The distance between the objective and the eyepiece can be adjusted so that the telescope forms a clear image at the cross wires.

The telescope is attached to a circular scale and both can be rotated together. The telescope and prism table are provided with radial screws for fixing them at a desired position and tangential screws for fine adjustments.

### *Preliminary adjustments of the spectrometer*

The following adjustments must be done in a spectrometer before doing the experiment.

*(a)* ***Adjustment of the eyepiece:*** The telescope is turned towards an illuminated surface and the eyepiece is moved to and fro until the cross wires are clearly seen.

*(b)* ***Adjustment of the telescope:*** The telescope is adjusted to receive parallel rays by focusing it to a distant object to get a clear image on the cross wire.

*(c)* ***Adjustment of the collimator:*** The telescope is brought in line with the collimator. The distance between the illuminated slit and the lens of the collimator is adjusted until a clear image of the slit is seen at the cross wire.

*(d)* ***Levelling of the prism table:*** The prism table is brought to the horizontal level by adjusting the levelling screws and it is ensured by using sprit level.



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